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Procedia Computer Science 259 (2025) 1773–1782

Procedia Computer Science

www.elsevier.com/locate/procedia

Sixth International Conference on Futuristic Trends in Networks and Computing Technologies (FTNCT06) held in Uttarakhand, India

Predicting Parkinson's Disease: An Explainable AI approach with Feature Analysis

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Abstract

Parkinson's disease (PD) is a degenerative neurological condition that significantly lowers a person's standard of living. Symptom management and better patient outcomes depend on early diagnosis and intervention. To forecast Parkinson's disease, this study looks into the use of various ML models, such as Decision Trees, XGBoost, AdaBoost etc. To improve model interpretability, the study highlights the significance of Explainable Artificial Intelligence (XAI) approaches. The essential elements that had a substantial impact on the models' predictions were found by using feature importance analysis. In addition to illuminating the disease's fundamental causes, this study offers medical professionals' practical insights to direct clinical judgment. The integration of machine learning with XAI techniques addresses the critical need for transparency in AI applications within healthcare, ensuring that practitioners can trust and understand the tools they use. Among the classifiers evaluated, the AdaBoost classifier demonstrated exceptional performance, achieving the highest accuracy of 98.34%. This result highlights the strong performance of the AdaBoost algorithm in accurately predicting Parkinson's disease, demonstrating its suitability and effectiveness for this application. These algorithms are computationally less expensive than the deep learning algorithms and hence could be efficiently used for real-time systems. Also in this paper the contribution of features have been mapped to the accuracy, which shows the explainability of features. The work aims to set the stage for future developments in neurodegenerative disease predictive modeling through thorough analysis and validation.

Keywords: ML Models; Parkinson's Disease; Feature Extraction; Explainable AI; Image Processing

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10.1016/j.procs.2025.04.132

1. Introduction

Neurological disorders are very critical and difficult to detect in their early stages. Parkinson's disease (PD) is one of them. Some of its characteristics affecting motor functions are trembling of hands and feet and stiffness. Other non-motor symptoms include mood swings, forgetfulness etc. [1]. With an estimated 10 million people afflicted globally, Parkinson's disease (PD) has a substantial cost to patients and healthcare systems. One of the major problems with this disease is its diagnosis and identification of symptoms [2]. Over the past few years this disease has shown significant growth, and its varying genotypes pose a challenge in its recognition. Parkinson disease is a brain dysfunctioning issue, generally observed in mid-age but now a days any age person might get affected. The main problem is that disease could not be treated with medicines so easily. Early diagnosis can significantly enhance quality of life and delay the disease's course; therefore, it is essential for successful care and intervention. This is the main motivation for this work. Conventional diagnostic techniques place a great deal of emphasis on subjective clinical evaluations and patient histories, which may cause diagnoses to be made later than necessary. New developments in machine learning (ML) present viable substitutes for Parkinson's disease early diagnosis and prediction [3]. Machine learning algorithms can find patterns and relationships in complex datasets that might not be visible using more traditional techniques. This capacity is especially helpful when it comes to Parkinson's disease (PD), as the initial symptoms can be mild and simple to miss. Not many works could be found in the literature, on use of simple ML models with variable feature extraction techniques for detection of Parkinson’s disease. This is the main research gap addressed. These days, machine learning is used for a wide range of different purposes in the medical field, including the detection of malarial infections [4], the prediction of heart disease [5], and many more.

In this study, various ML classifiers used to predict the onset of Parkinson's disease. The effectiveness of these models is evaluated based on their predictive accuracy and interpretability. Furthermore, the integration of Explainable Artificial Intelligence (XAI) techniques is a focal point of this research. XAI enhances the transparency of machine learning models, allowing healthcare professionals to understand the rationale behind predictions [6]. By employing feature importance analysis, this study aims to uncover the key factors influencing model predictions, thereby providing actionable insights for clinical practice. The main objective of this study is to achieve better accuracy through low cost ML models. Through rigorous analysis and validation, the study seeks to pave the way for future advancements in predictive modeling for neurodegenerative diseases, ultimately aiming to improve patient outcomes and enhance the quality of care. Subsequent sections of paper include the following: the second section ‘Literature Review’ discussed about some of the recent related works; the third section ‘Materials and Methods’ discussed various technologies and dataset used; the fourth section ‘Result and Discussion’ discussed the study’s findings and result; in the fifth section ‘Conclusion and future work’ followed by References.

2. Literature Review

In 2023, Chandrasekaran et al. [7] developed a KNN-based evolutionary instance-based technique for classification of Parkinson's disease (PD) patients. They incorporated deep learning models, such as Neural Networks (NN), Artificial Neural Networks (ANN), and Recurrent Neural Networks (RNN), to achieve the desired outcomes. The proposed system also utilized fuzzy logic as an intelligent approach. The method automatically determined two critical parameters through FKNN optimization combined with chaos theory. This new FKNN method outperformed other ML models such as—Support Vector Machine (SVM), LOGO-SVM, and Kernel Extreme Learning Machine (KELM)—as well as five nature-inspired FKNN models. Simulation results indicated that the FKNN approach could serve as an effective computer-aided diagnostic tool for medical decision-making, with the proposed model achieving a 99.2% accuracy rate in predicting the disease when tested on the data. In 2023, Alshammri et al. [8] focused on developing an effective method for detecting Parkinson's disease (PD) using voice samples. UCI dataset was used, which included 195 voice signal recordings from 48 healthy controls (HC) and 147 people with Parkinson's disease (PD). Numerous deep learning and conventional machine learning techniques were used in the investigation. In cooperation with the National Center for Voice, the University of Oxford (UO) repository provided the voice data for the study. Among the algorithms tested, the MLP demonstrated the highest performance, achieving an overall accuracy of 98.31%, a recall of 98%, a precision of 100%, and an F1-score of 99%.

Mall et al. [9] (2022) introduced a novel Vote Count Model (VCM) for evaluating and classifying Parkinson's disease. The study incorporated data preprocessing techniques to optimize the datasets for improved classification performance. From the results it could be inferred that gave better performance than existing machine learning algorithms, including Decision Tree, SVM, KNN, Random Forest, and XGBoost. The ensemble model they developed proved superior to these traditional methods, achieving an accuracy of 94.87%, a Matthews Correlation Coefficient (MCC) of 81.99%, and an F1 score of 94.52%. Krishna et al. [10] (2021) proposed a Machine Learning (ML)-driven Logistic Decision Regression (LDR) algorithm to address the complex task of classifying Parkinson's disease. Early detection of Parkinson's is crucial for better disease management and improved patient care, ultimately enhancing the quality of life. To support this goal, the researchers developed and introduced an automatic prediction system based on machine learning. The LDR algorithm demonstrated strong performance, achieving a sensitivity of 95%, a precision of 93%, and an overall classification efficiency of 97%.

In their study, Salmanpour et al. [11] (2021) focused on assessing Parkinson's disease (PD) by tackling both the identification of PD subtypes (an unsupervised task) and the prediction of these subtypes (a supervised task). They developed and applied two distinct hybrid machine learning systems (HMLSs) to determine the optimal combinations for these tasks: (i) unsupervised clustering for PD subtype identification and (ii) supervised classification for predicting these subtypes after four years. The initial section of the study employed feature selection algorithms (FSAs) to rank the features prior to clustering, then HMLSs were used to identify the best pairings, which included FSASL + KMA and UDFSA + KMA. Radiomics features were the most important ones found. The study found that PD patients' outcomes can be consistently predicted by combining non-imaging data with SPECT-based radiomics features and using HMLSs efficiently. In the prediction task, their approach achieved an accuracy exceeding 90%. Study conducted by Nilashi et al. [12] (2022) for remotely tracking the progression of Parkinson's disease employed the Expectation-Maximization (EM) and Self-Organizing Maps (SOM) algorithms to create multiple segments, which were then used to form ensembles for final clustering outcomes. The comparison of different clustering and prediction methods was the main goal of the study. The results showed that, with the maximum R2 score of 91%, the HGPA + EM + SVR ensemble obtained greater accuracy.

In their 2023 paper, Govindu et al. [13] used machine learning techniques applied to audio recordings from people with Parkinson's (PWP). This innovative approach highlighted the potential of audio as a non-invasive biomarker for detecting PD. The performance of the SVM model, which attained an accuracy of 91.836% and a sensitivity of 0.94, was also highlighted by the study. Senturk et al. [14] (2020) used voice signal analysis from both PD patients and healthy individuals to create a feature selection (FS) based decision support system intended for diagnosis of Parkinson's disease (PD). To improve model performance and accuracy and lower the computing cost of the classification task, the study combined a variety of FS techniques with different classification algorithms. Both with and without the use of FS, the classification systems' accuracy was evaluated, indicating the substantial influence of FS on the outcomes. By demonstrating that their early diagnosis approach could accurately identify Parkinson's disease (PD) even at an early stage, the researchers may be able to slow the disease's course. Remarkably, they achieved an accuracy of 93.84% using a minimal number of voice features for Parkinson's diagnosis.

In their study, Wang et al. [15] (2020) proposed a deep learning framework aimed at diagnosis of PD. The model was created with the intention of automatically differentiating between those without Parkinson's disease (PD) and those who have it. It focuses on premotor characteristics such olfactory loss, sleep behavior disturbance, and rapid eye movement (REM). With 96.45% accuracy, the deep learning model showed good detecting capabilities. In their 2024 study, Kumar et al. [16] aimed to utilizing audio samples, to determine the best machine learning model for Parkinson's disease early detection. The methodology involved collecting audio data on voice modulations from Parkinson's patients, sourced from the PPMI and UCI databases. 75% of the dataset was used to train four ML models. The Logistic Regression model produced an accuracy of 85.71% and a sensitivity of 0.89 for the categorization of Parkinson's disease using vowel phonation data. This study aims at exploring novel feature extraction techniques and thus improving the generalization of machine learning models for automated detection of Parkinson's disease.

3. Materials and Methods

The methodological framework applied for Parkinson's disease detection involves a multi-step approach for the effective diagnosis of the disease. It involves taking relevant information out of x-ray images using various methods for extracting features and applying different machine learning classifiers subsequently. Details about the feature extraction methods and classifiers for machine learning used in this investigation, the characteristics of the dataset used for learning, and the metrics utilized to evaluate the quality of the generated images have been explained in this section.

3.1. Feature Extraction Techniques

- Local Binary Pattern (LBP)

Local Binary Pattern (LBP) is a widely used feature extraction technique in image processing, particularly valued for its effectiveness in texture analysis [17]. This method operates by examining each pixel in an image relative to its surrounding neighborhood. For each pixel, LBP generates a binary code based on whether the values of neighboring pixels are greater than or less than the central pixel value. Specifically, the central pixel is compared to its surrounding eight pixels, and a binary code is created by assigning a '1' for pixels with higher values and a '0' for those with lower values. This binary code is then converted into a decimal number, which is used to construct a histogram that reflects the texture information of the image. The working of LBP is depicted in Fig. 1.

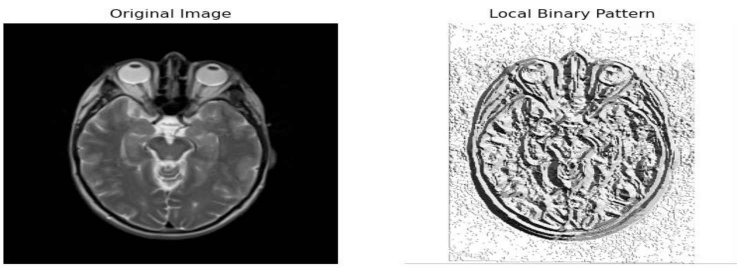

Fig. 1. Working of Local Binary Pattern.

LBP is notable for its simplicity and computational efficiency, making it highly suitable for real-time applications. Its robustness to variations in illumination and its ability to capture local texture details effectively have made it a popular choice in a range of applications, including face recognition, image retrieval, and medical image analysis. The technique's capacity to encode texture patterns into a compact and descriptive histogram enables it to perform well even with relatively low computational resources, making it an indispensable tool in many image processing tasks.

- Local Directional Pattern (LDP)

Local Directional Pattern (LDP) is a sophisticated technique for feature extraction used in texture analysis within image processing [18]. Unlike conventional methods that concentrate on pixel intensity alone, LDP focuses on capturing texture information through the analysis of gradient directions around each pixel. The process begins by calculating the gradient magnitude and direction at each pixel using edge detection methods such as Sobel filters. These gradients are then employed to define local directional patterns in relation to the central pixel. These patterns are encoded in binary format to reflect various directional orientations in the pixel's neighborhood. In constructing a Local Directional Pattern, the image is segmented into small blocks or regions. Within each block, the gradient directions are evaluated, and each pixel receives a binary code based on its directional relation to its neighbors. This binary code is subsequently transformed into a decimal number, creating a histogram that represents the texture characteristics of the image. LDP improves the depiction of local texture patterns by emphasizing directional information over mere intensity differences. This enhancement makes LDP more resilient to changes in lighting and noise. It is especially valuable in tasks such as image classification, object recognition, and medical imaging, where precise directional features are crucial for increasing the accuracy and effectiveness of analysis. Fig. 2 depicts the working of LDP.

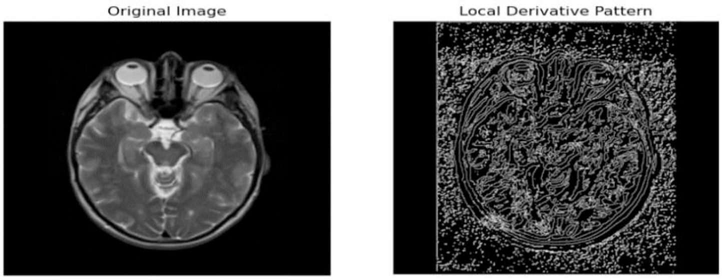

Fig. 2. Working of Local Directional Pattern.

- Discrete Wavelet Transform (DWT)

The Discrete Wavelet Transform (DWT) is used for analyzing signals and images at multiple scales and resolutions [19]. Unlike the Fourier Transform, which provides only frequency information, DWT captures both frequency and spatial details, making it ideal for non-stationary signals and multi-resolution analysis. DWT decomposes an image into a series of wavelets—oscillatory functions of varying scale and position. This decomposition is repeated to produce a multi-level representation. The result includes coefficients organized into four sub-bands: approximation (low frequency), and detailed coefficients (horizontal, vertical, diagonal high frequencies). This enables efficient data compression, noise reduction, and feature extraction. DWT is widely used in image processing for tasks such as compression, denoising, and texture analysis. DWT is represented in Fig. 3.

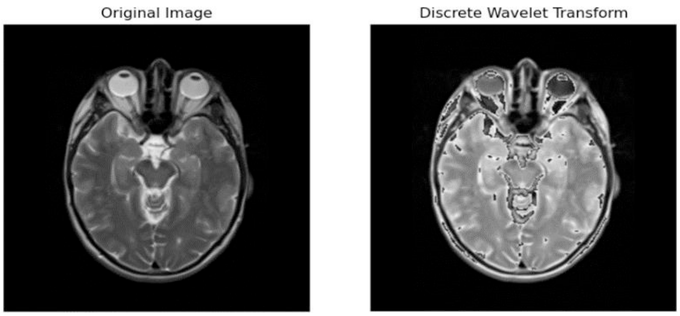

Fig. 3. Working of Discrete Wavelet Transform.

- Hough Transform

The Hough Transform is an image processing technique used to identify geometric shapes, such as lines, circles, and ellipses, within an image [20]. Shapes are represented as curves or points in the parameter space created by converting image space. In line detection, the angle and distance of each line from the origin are represented by parameters, and the Hough Transform transfers each edge point in the image to a sinusoidal curve in the parameter space. Intersections of these curves reveal the presence of lines. For circle detection, the transform projects each edge point into a three-dimensional parameter space where the parameters define the circle's center and radius. Accumulation of votes in this space identifies potential circles. The Hough Transform is effective in dealing with noise and can detect shapes even if they are partially obscured. It is commonly used in fields such as image analysis, computer vision, and automated inspection to detect and recognize geometric patterns in images.

- Gabor Transform

The Gabor Transform is a method employed in image processing and computer vision to analyze textures and extract features [21]. It utilizes Gabor filters, which are sinusoidal wave functions modulated by a Gaussian envelope, to process images. These filters are adept at capturing specific spatial frequencies and orientations, which makes them highly effective for texture and pattern analysis. Gabor filters are available in various orientations and scales, enabling them to identify different texture characteristics. Applying these filters to an image produces a series of filtered result

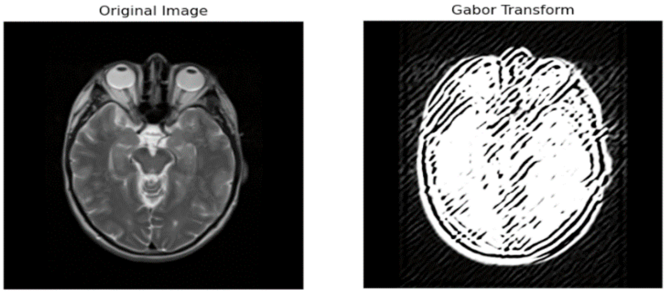

Fig. 4. Working of Gabor Transform.

that emphasize different textures and patterns. These filtered images are then used to analyze and describe textures, detect edges, or identify specific patterns within the image. Gabor Transform is depicted in Fig. 4. The Gabor Transform is particularly useful for its multi-scale and multi-orientation capabilities, which are valuable in tasks such as face recognition, fingerprint analysis, and texture classification. Its ability to capture detailed spatial frequency information makes it a robust tool for analyzing intricate textures and patterns in images.

- Discrete Fourier Transform (DFT)

The Discrete Fourier Transform (DFT) is a method for analyzing the frequency components of discrete signals or images [22]. It transforms a sequence of data into its frequency-domain representation by breaking the signal down into sinusoidal functions with distinct frequencies, amplitudes, and phases. This process produces a set of coefficients that describe the frequency characteristics of the original data as shown in Fig.5.

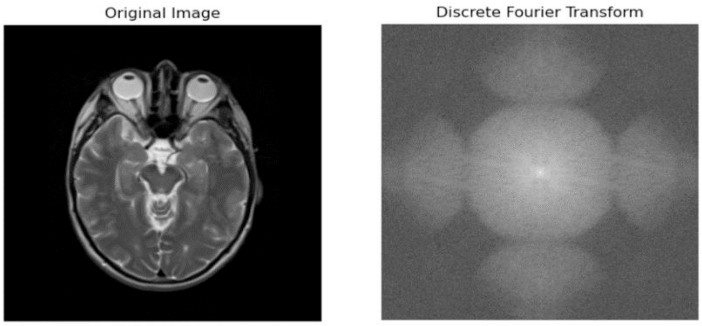

Fig. 5. Working of Discrete Fourier Transform.

In image processing, DFT is used to examine and modify frequency information, facilitating operations like filtering, compression, and feature extraction. Its capability to deliver a detailed frequency-domain view makes it a crucial tool for in-depth signal and image analysis. Fig. 1 to 5 are generated during simulation of the ML models.

3.2. Machine Learning Models

- XG-Boost

XGBoost (Extreme Gradient Boosting) [23] has many decision trees, with each tree trying to correct the mistakes committed by the one before it. XGBoost enhances speed and efficiency through techniques such as regularization and parallel processing, making it well-suited for large datasets and intricate models.

- K-Nearest Neighbors

The K-Nearest Neighbors (KNN) algorithm [24] determines the k closest data points, using a distance metric such as Euclidean distance, to a given query point. While KNN is appreciated for its straightforward approach and effectiveness, it can become computationally demanding with large datasets.

- Support Vector Classifier

The method by which Support Vector Classification (SVC) partitions data into discrete classes and maximizes the margin between them is locating the ideal hyperplane. SVC is renowned for its accuracy and durability in high-dimensional areas, making it a particularly effective method for challenging classification tasks.

- Decision Tree

A Decision Tree by making a sequence of judgments depending on feature values of subset trees. A decision criterion is represented by each node in the tree, while leaf nodes provide the final prediction. Decision Trees are appreciated for their straightforwardness and ease of interpretation, efficiently handling both numerical and categorical data.

- Naïve Bayes

Naive Bayes estimates class probabilities using prior knowledge and the likelihood of features. Its simplicity and effectiveness make it particularly useful for large datasets and text classification tasks. It uses Bayes Theorem.

- AdaBoost

AdaBoost (Adaptive Boosting) is an algorithm that enhances weak classifiers by focusing on the errors made by previous models. It adjusts the weights of misclassified instances to improve accuracy, thereby creating a more robust and effective classifier [25].

- CatBoost

CatBoost is a gradient boosting algorithm tailored for handling categorical data [26]. It builds decision trees in sequence to improve accuracy, providing strong performance and efficiency while minimizing the need for extensive preprocessing of both categorical and numerical features.

- Random Forest (RF)

Random Forest is a technique where multiple trees are used to increase performance. By combining the results of different trees, it reduces overfitting and boosts reliability. A random sampling of the data and features is used to train each tree, and the final prediction is determined by average for regression or majority voting for classification.

- Gradient Boosting (GB)

Gradient Boosting is an ensemble learning method that creates a series of models in sequence, where each model aims to correct the mistakes made by its predecessors [27]. By focusing on the residual errors of the aggregated predictions, this technique enhances overall accuracy and reduces errors through iterative refinement.

- Logistic Regression (LR)

A logistic function is used in the classification model known as logistic regression to determine the likelihood that an input will belong to a specific class. It returns a number, ranging from 0 to 1, indicating the probability that the input will fall into one of the two available categories. This approach is valued for its ease of use and efficiency in jobs involving binary categorization.

3.3. Dataset Description

The study utilizes a comprehensive dataset from Kaggle for the Parkinson's Disease prediction with the help of classification models [28]. The dataset named 'Parkinson's disease_FMRI+Images' consisted of 606 Functional MRI Brain Scan Images.

- Data Collection and Preprocessing

The datasets utilized in this study were obtained from Kaggle. To maintain consistency and ensure uniform input dimensions, all images were resized to 128 by 128 pixels. Additionally, pixel values were normalized as part of the preprocessing process. Images were augmented using techniques like flipping, rotation, and adding noise to increase the diversity of the dataset.

- Feature Extraction

Features were retrieved from the images using the feature extraction techniques: LDP, LBP, DWT, Hough Transform, Gabor Transform and Discrete Fourier Transform after preprocessing and grayscale conversion was done to capture the important properties required for successful classification. The machine learning models were subsequently trained using these extracted attributes. The extracted features' histograms are generated and normalized.

- Dimensionality Reduction and Standardization

The dimensionality of the characteristics was decreased through the application of Principal Component Analysis (PCA). Standard Scaler is used to standardize features. Next, the dataset is divided into testing and training sets. On the testing set, a number of classifiers are trained and assessed. Each classifier's accuracy is calculated and reported. The ML algorithms used have been referred from the literature review done for this research work.

4. Results and Discussion

Evaluating the outcomes is critical to identify the optimal classifier. The primary performance metric used for evaluating the machine learning models was Accuracy. The results indicated that AdaBoost classifier outperformed all the other models. The reason may be cumulative accuracy of Adaboost that it acquires from different ML models. Random Forest achieved an accuracy of 98.34%. So Random Forest though achieved a bit less accuracy than Adaboost could be treated as lesser cost model as it did not get the cumulative accuracy of other ML models. After AdaBoost, XG Boost and Logistic Regression also performed relatively well with an accuracy of 97.93%. Analysis of different bin sizes in the Greyscale color space revealed that the optimal performance occurred with bin sizes of 8. This is a good indication. More bin sizes means more time complexity. But this research finds a good accuracy. Table 1. shows the accuracy of the ML classifiers in different bin sizes. Getting maximum accuracy in bin size 8 defies the common notion of more features, more accuracy. Using bin size 8 reduces the feature space largely and hence computationally less expensive. This also shows the principle of overfitting with a higher number of features. Performance analysis is graphically shown in Fig.6.

Table 1. Performance evaluation of ML models

Models	8 Bins	16 Bins	32 bins	64 bins	128 bins	256 bins
XGBoost	97.93	97.93	97.10	96.27	97.51	97.51
KNN	96.68	96.27	92.95	90.04	96.27	90.04
SVC	97.10	96.27	97.10	92.95	97.51	94.61
Decision tree	95.44	94.19	95.85	92.53	95.02	95.85
Naive Bayes	81.74	22.84	34.02	78.42	90.04	87.55
AdaBoost	98.34	97.51	97.10	94.19	96.27	96.27
CatBoost	97.51	97.93	97.10	95.85	97.93	97.51
LR	97.93	96.68	96.27	95.44	96.27	94.61
GB	96.27	96.27	97.51	95.44	97.93	95.85
RF	97.10	97.93	97.10	95.85	96.68	97.51

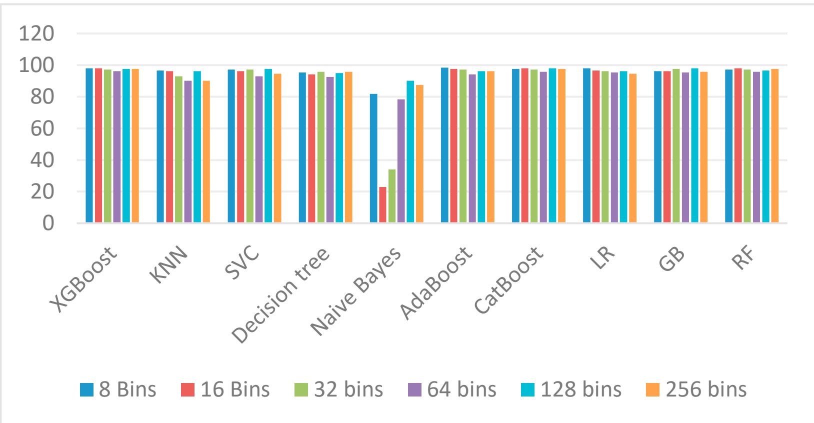

Fig. 6. Comparative performance analysis of ML models

After that, explainable AI techniques were used to learn more about the model's decision-making methodology. Feature importance analysis was conducted to determine and quantify the most significant features influencing the predictions. This method enhanced understanding of the model's behavior, promoting transparency and trust in the results, and enabling more informed decisions based on the primary factors driving the outcomes. Fig 7. represents the cumulative variance explained by the principal components derived from PCA. As from the experimentation it is found that, the model achieved best accuracy with all the 14 components, so all the features were used for evaluation.

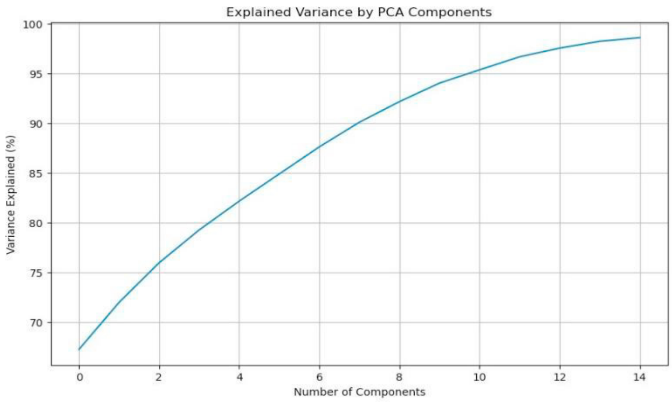

Fig. 7. Cumulative Explained Variance by PCA Components.

The y-axis in the above graph shows the percentage of variance, and the x-axis shows the number of primary components. explained by those components. As the number of components increases, a greater portion of the dataset's variance is captured. The initial few components account for a significant amount of variance, as seen by the steep increase in the curve. However, the curve begins to plateau after approximately 10 components, suggesting that adding more components beyond this point results in diminishing returns in terms of capturing additional variance. This plateau indicates that most of the relevant information in the dataset can be captured using around 10 components. By selecting this number of components, one can achieve a balance between reducing dimensionality and preserving the majority of the dataset's information, which is crucial for effective feature reduction in machine learning models. Overall, this graph demonstrates the efficacy of PCA in selecting better or optimal feature set while retaining the key information necessary for further analysis.

5. Conclusion and Future work

This study evaluated the effectiveness of various machine learning classifiers, including AdaBoost, XGBoost, Random Forest, and others, for predicting Parkinson's disease. Complex patterns in the image data were extracted using advanced feature extraction techniques including Discrete Wavelet Transform (DWT), Local Binary Patterns (LBP), Local Directional Patterns (LDP), and others. Among the classifiers tested, AdaBoost demonstrated superior performance, achieving an accuracy of 98.34%. Explainable AI approaches have been used, especially feature importance analysis, to learn more about the models' decision-making processes. This analysis identified the most influential features, ensuring transparency and trustworthiness in the results. Additionally, the use of Principal Component Analysis (PCA) was explored, with findings suggesting that around 10 components were optimal for retaining most of the relevant information while minimizing model complexity.

To improve the generalizability of the models, future research should concentrate on growing the dataset to include a wider spectrum of Parkinson's disease cases. Exploring more advanced feature extraction methods and deep learning approaches, could further improve classification accuracy. Finally, applying these techniques in real-world clinical settings could lead to the development of a robust, interpretable tool for early Parkinson's disease detection, aiding healthcare professionals in making more informed decisions.

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